



AEF Crop Intelligence: User Guide

AEF Crop Intelligence is a decision support tool for farmers and agronomists. It combines a **Digital Twin** of your field (a computer simulation of crop growth) with **Satellite Intelligence** (real-time observations from space) to help you optimize yield, manage diseases, and reduce input costs.

It answers three critical questions:

1. **How** is my crop doing compared to its **potential**?
2. **Where exactly** are the **disease/stress hotspots**?
3. **What actions** should I take?



Getting Started

The app guides you through a 5-step configuration process to build your field's Digital Twin.

Step 1: 🌍 Geography

Define exactly where your field is located.

1. **Search:** Type your nearest town (e.g., "Garoua, Cameroon") and click 🔍 `Locate`.
2. **Draw:** Use the Polygon tool (pentagon icon) on the left of the map to trace your field boundaries.
 - **Validation:** The system uses satellite data (ESA WorldCover) to check if you are drawing on valid farmland. It will reject polygons drawn on water, forests, or cities.
3. **Confirm:** Once valid, you will see a green success message with the field size (in hectares). Click `Next Step` ➡.

Step 2: 🌱 Crop System

Tell the system what you are growing.

1. **Select Crop:** Choose from major crops like Maize, Cassava, Cocoa, etc.
2. **Variety:** Select the specific variety (e.g., "Drought Tolerant" vs "High Yield").
 - *Note:* The app automatically suggests a standard **Planting Density** (plants/ha) based on the crop, but you can adjust this if you plant differently.
3. **Dates:** Set your planting date.

Step 3: 🦟 Disease Surveillance

Input any observations you have made on the ground.

1. **Select Threat:** Choose the primary disease you are worried about (e.g., "Cassava Mosaic Disease").
2. **Mark Spots:** Click on the map where you have seen infected plants.

3. **Set Severity:** In the table on the right, enter the number of infected plants observed at each spot.
 - *Why?* The simulation uses this as "Patient Zero" to model how the disease will spread across your field over time, helped by automatically computed environmental risk.

Step 4: Soil & Management

Configure the soil to ensure the water balance simulation is accurate.

1. **Auto-Detect:** Click  Auto-Detect Soil Profile. The app will query global soil databases (OpenLandMap) to find the texture (Sand/Loam/Clay) and organic carbon levels at your specific GPS coordinates.
2. **Manual/Expert Mode:** If you have lab results, you can toggle "Expert Mode" to manually edit soil layers and hydraulic properties.
3. **Management:** Use the tables to schedule **Irrigation** (mm) and **Fertilizer** (kg N/ha) events.

Step 5: Launch

Review your summary and click  **Initialize Digital Twin**. This triggers the simulation engine (most of time, you'll need to switch to the Dashboard Intelligence tab afterwards).

Intelligence Dashboard

Once initialized, you are taken to the Dashboard. This is your command centre.

1. Seasonal Timeline

Use the **slider** at the top to travel through time.

- **Past:** See historical growth and stress.
- **Future:** See the predicted state of your crop for the rest of the season.

2. Spatial Maps

Visualizes the field status for the selected date.

- **Disease Severity:** Red zones indicate high infection risk. Use this to direct your scouting or spraying.
- **Yield Potential:** Green zones show where biomass is accumulating well.

3. Daily Dynamics

These charts show the heartbeat of your crop.

- **LAI & Biomass:** Tracking growth.

- **Soil Water & Nmin:** Monitoring the "fuel" in the soil. Blue line = Water, Orange line = Nitrogen.
- **Reality Check (NDVI):** This is the most powerful tab.
 - **Green Line:** What the computer *thinks* your crop looks like.
 - **Blue Dots:** What the *satellite actually sees*.
 - **Interpretation:** If the Blue Dots are significantly lower than the Green Line, something is wrong (pests, unmodelled disease, or flooding) that the model didn't predict.



The Report

Go to the **Report** tab to generate a formal PDF dossier.

- **Executive Summary:** Total estimated production in tonnes.
- **Diagnostics:** Explains *why* yield was lost (e.g., "20% loss due to Nitrogen stress in Week 4").
- **Recommendations:** specific, actionable advice (e.g., "Apply side-dressing," "Scout Sector B").